

Angles & Angle Relationships

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Standard: CCSS.MATH.CONTENT.7.G.B.5 — Use facts about supplementary, complementary, vertical, and adjacent angles in a multi-step problem to write and solve simple equations for an unknown angle.

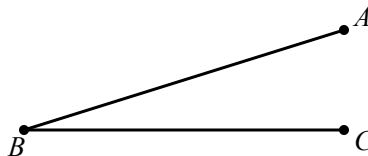
Part A — Vocabulary

Directions: Match each term in the table to its correct definition. Write the letter (A–E) on the line.

Term	Definition (choose one: A, B, C, D, or E)
1a. Point	Answer: _____
1b. Line Segment	Answer: _____
1c. Ray	Answer: _____
1d. Vertex	Answer: _____
1e. Angle	Answer: _____

- A. A figure formed by two rays that share a common endpoint.
- B. A part of a line that has two endpoints.
- C. A location with no length, width, or height.
- D. The point where two lines or segments meet.
- E. A part of a line that starts at one point and extends forever in one direction.

2. The figure below shows an angle with vertex B and rays $\overrightarrow{BA}$ and $\overrightarrow{BC}$. Write **three different names** for this angle.



(i) _____ (ii) _____ (iii) _____

Part B — Classify the Angle

Directions: *Classify each angle as acute, right, obtuse, or straight.*

3. $\angle 1$ measures 35° . Classification: _____

4. $\angle 2$ measures 90° . Classification: _____

5. $\angle 3$ measures 142° . Classification: _____

6. $\angle 4$ measures 180° . Classification: _____

Part C — Angle Relationships

Directions: Use what you know about complementary, supplementary, vertical, and adjacent angles. Show your work.

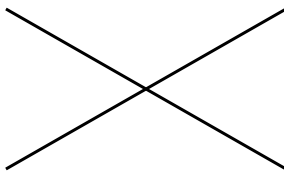
7. Two angles are **complementary**. One angle measures 27° . What is the measure of the other angle?

Answer: _____

8. Two angles are **supplementary**. One angle measures 118° . What is the measure of the other angle?

Answer: _____

9. $\angle 1$ and $\angle 2$ are **vertical angles**. If $m\angle 1 = 64^\circ$, find $m\angle 2$. Briefly explain why.



$$m\angle 2 = \underline{\hspace{2cm}}$$

Why?

10. $\angle ABC$ and $\angle CBD$ are **adjacent angles** that together form a **straight angle**. If $m\angle ABC = 75^\circ$, find $m\angle CBD$.

$$m\angle CBD = \underline{\hspace{2cm}}$$